System Requirements

"The following requirements are derived from the system use case diagram and project objectives to ensure complete system coverage".

Functional Requirements:

Based on the system use case diagram, the Pharmacy Management System shall provide the following functionalities:

1. User Account Management

- The system shall allow customers to register a new account.
- The system shall allow users (customers and admins) to log in securely.
- The system shall allow admins to manage user accounts.

2. Medicine Browsing and Searching

- The system shall allow customers to view available medicines.
- The system shall display medicine details such as name, price, and availability.
- The system shall allow customers to search for medicines by name or category.

3. Order Management

- The system shall allow customers to place orders for medicines.

- The system shall allow customers to view their orders.
- The system shall allow pharmacists to approve orders.
- The system shall process orders and calculate total cost.

4. Prescription Handling

- The system shall allow pharmacists to view prescriptions.
- The system shall support prescription verification before order approval.

5. Medicine Management (Admin)

- The system shall allow admins to add new medicines.
- The system shall allow admins to update medicine information.
- The system shall allow admins to delete medicines.

6. Inventory Management

- The system shall allow pharmacists to manage inventory.
- The system shall track stock levels in real time.
- The system shall allow updating stock quantities.

7. Reporting

- The system shall allow admins to view reports.
- The system shall generate sales and financial reports.

Non-Functional Requirements:

The system shall satisfy the following non-functional requirements:

1. Performance

- The system shall respond within 2–3 seconds for user actions.
- The system shall handle multiple users simultaneously.

2. Security

- The system shall ensure secure login using authentication mechanisms.
- The system shall enforce role-based access control (Customer, Admin, Pharmacist).
- The system shall protect sensitive data from unauthorized access.

3. Usability

- The system shall provide a simple and user-friendly interface.
- The system shall be easy to learn for pharmacy staff and customers.

4. Availability

- The system shall be available during working hours with minimal downtime.
- The system shall run on a local network.

5. Compatibility

- The system shall be compatible with barcode scanners and receipt printers.
- The system shall integrate with external drug databases.

6. Maintainability

- The system shall be easy to maintain and update.
- The system shall support modular development.

7. Scalability

- The system shall support future expansion (e.g., multiple branches).

8. Reliability

- The system shall ensure accurate order processing.
- The system shall prevent data loss and maintain data integrity.